# Algebra Qual Fall 2018

## Problem 1

(10 points) Let $G$ be a finite group whose order is divisible by a prime number $p$. Let $P$ be a normal $p$-subgroup of $G$ (so $|P| = p^c$ for some $c$).

**(a)** Show that $P$ is contained in every Sylow $p$-subgroup of $G$.

**(b)** Let $M$ be a maximal proper subgroup of $G$. Show that either $P \subseteq M$ or $|G/M| = p^b$ for some $b \leq c$.

*Proof.* **(a)** Write $|G| = mp^n$ where $p \nmid m$, with $|P| = p^k$. If $k = n$ then $P$ is a sylow $p$-subgroup and its normality implies it is the unique one since all sylow $p$-subgroups are conjugate.

Now suppose $k < n$ and consider the quotient $G/P$. This has order $mp^{n-k}$. Since sylow subgroups always exist there is a sylow $p$ subgroup $P' \leq G/P$ and this group has order $p^{n-k}$ since thats the largest power of $p$ dividing $G/P$. Now let $q : G \to G/P$ be the quotient. Then

$$\left|q^{-1}\left(P'\right)\right| = \left|P'\right| \cdot |P| = p^n$$

so by the lattice theorem $Q := q^{-1}\left(P'\right)$ is a sylow $p$-subgroup containing $P$. Now by normality of $P$

$$P = gPg^{-1} \subseteq gQg^{-1}, \text{ for all } g \in G.$$

Then since sylow subgroups are conjugate $gQg^{-1}$ is an arbitrary sylow $p$-subgroup thus $P$ is contained in the intersection of all of them.

**(b)** Suppose that $P \not\subseteq M$. Then as $P$ is normal $MP$ is a group, and $M$ is properly contained in $MP$ since there is a $p \in P \setminus M$ and $p \in MP$. Then by maximality of $M$ we see that $MP = G$. Then

$$G/M = MP/M = P/M, \text{ as sets of cosets, not as quotient groups.}$$

In otherwords every coset $gM$ has a representative in $P$, and $|G/M| = |P/M|$.

Now let $P$ act on $P/M$ by left multiplication. Note that this action is transitive since for example $p'p^{-1} \cdot pM = p'M$. Then the orbit stabilizer theorem yields

$$|P| = |P/M| \cdot |\text{Stab}(M)|.$$

Then $\text{Stab}(M) = P \cap M$ since $pM = M$ iff $p \in M$. Then $|P \cap M|$ is a proper subgroup of $P$ since $P$ is not contained in $M$ thus $|P \cap M| = p^k$ for $k < c$ and

$$|P/M| = p^{c-k}$$

where since $k \geq 0$ we have $c - k \leq c$. □

## Problem 2

(10 points)

**(a)** Suppose the group $G$ acts on the set $X$. Show that the stabilizers of elements in the same orbit are conjugate.

**(b)** Let $G$ be a finite group and let $H$ be a proper subgroup. Show that the union of the conjugates of $H$ is strictly smaller than $G$, i.e.

$$\bigcup_{g \in G} gHg^{-1} \subsetneq G.$$

**(c)** Suppose $G$ is a finite group acting transitively on a set $S$ with at least 2 elements. Show that there is an element of $G$ with no fixed points in $S$.

*Proof.* **(a)** Suppose $x, y \in O_x$. That is there is $g \in G$ such that $g \cdot x = y$. Let $F(x)$ denote the stabilizer of $x$. We claim

$$gF(x)g^{-1} = F(y).$$

Indeed suppose $h \in F(x)$. Then

$$ghg^{-1} \cdot y = gh \cdot \left(g^{-1} \cdot y\right) = gh \cdot x = g \cdot (h \cdot x) = g \cdot x = y$$

so $gF(x)g^{-1} \subseteq F(y)$. Now to see the map is surjective suppose $h \in F(y)$. We claim $g^{-1}hg \in F(x)$ and thus its image under $g(\cdot)g^{-1}$ is precisely $h$. We check the same equality

$$g^{-1}hg \cdot x = g^{-1}h \cdot y = g^{-1} \cdot y = x$$

as desired.

**(b)** This about the size of the union of conjugate subgroups being strictly smaller then the union of cosets because they have an intersection.

Let $g_1H, \ldots, g_kH$ be the distinct cosets of $H$ i.e. $k = [G : H]$. Then the distinct conjugate subgroups are $g_1Hg_1^{-1}, \ldots, g_kHg_k^{-1}$. Then

$$\begin{aligned}\left|\bigcup_{i=1}^{k} g_iHg_i^{-1}\right| &= \left|\{e\} \cup \bigcup_{i=1}^{k} g_iHg_i^{-1} \setminus \{e\}\right| \\ &\leq 1 + \sum_{i=1}^{k} (|H| - 1) = 1 + k\,|H| - k \\ &= |G| - k + 1\end{aligned}$$

since the identity lands in every conjugate subgroup. Since $H$ is proper $k \geq 2$ so the above quantity is less then $G$.

**Remark.** To formalize

$$(\#\text{conjugate subgroups of } H) \leq (\#\text{cosets of } H)$$

show that the map $gH \mapsto gHg^{-1}$ is well defined. It obviously surjective so we get the right inequality.

**(c)** This is straight from burnsides lemma. Recall $|S/G|$ is the number of orbits under the action of $G$. Then we have

$$1 = |S/G| = \frac{1}{|G|} \sum_{g \in G} |S^g|\,.$$

Then $|G| = \sum_{g \in G} |S^g|$. Now note that $S^e = S$ so we may write

$$|G| = |S| + \sum_{g \neq e} |S^g| \geq |S| + |G| - 1$$

so $|S| \leq 1$ a contradiction. □

## Problem 3

(10 points) Let $F \subset K \subset L$ be finite degree field extensions. For each of the following assertions, give a proof or a counterexample.

**(a)** If $L/F$ is Galois, then so is $K/F$.

**(b)** If $L/F$ is Galois, then so is $L/K$.

**(c)** If $K/F$ and $L/K$ are both Galois, then so is $L/F$.

*Proof.* **(a)** False for example if $L$ is the splitting field for $x^4 - 5$ (which is irred over $\mathbb{Q}$ by eisensteins with $p = 5$) then $\mathbb{Q}\left(5^{1/4}\right)$ is a subfield, but it is not Galois since it is not normal. It contains one root of $x^4 - 5$ but not all.

**(b)** This is true. Let $f(x) \in K[x]$ be the minimal polynomial of some $\alpha \in L$. Then since $L/F$ is Galois (algebraic since the extension is finite) there is a minimal polynomial over $F$ say $g(x) \in F[x]$. Now $g(x)$ is still an annihilating polynomial in $K[x]$ so by minimality $f(x) \mid g(x)$. Then since $g(x)$ splits in $L$ so does $f(x)$, and since $g(x)$ has distinct roots in $L$ so does $f(x)$.

**(c)** No. Consider the extensions

$$\mathbb{Q} \subseteq \mathbb{Q}\left(\sqrt{2}\right) \subseteq \mathbb{Q}\left(\sqrt[4]{2}\right).$$

$\mathbb{Q}\left(\sqrt{2}\right)$ is Galois over $\mathbb{Q}$ since its the splitting for the irreducible poly $x^2 - 2 \in \mathbb{Q}[x]$. Also $\mathbb{Q}\left(\sqrt[4]{2}\right)$ is Galois over $\mathbb{Q}\left(\sqrt{2}\right)$ since it is the splitting field for $\left(x^2 - \sqrt{2}\right) \in \mathbb{Q}\left(\sqrt{2}\right)[x]$, but $\mathbb{Q}\left(\sqrt[4]{2}\right)$ is not Galois over $\mathbb{Q}$ because it isn't normal. It contains one root of $x^4 - 2$, but not all. □

## Problem 4

(10 points) Let $V$ be a finite dimensional vector space over a field (the field is not necessarily algebraically closed). Let $\phi : V \to V$ be a linear transformation. Prove that there exists a decomposition of $V$ as $V = U \oplus W$, where $U$ and $W$ are $\phi$-invariant subspaces of $V$, $\phi|_U$ is nilpotent, and $\phi|_W$ is nonsingular.

*Proof.*

### Attempt

Consider the exact sequence

$$0 \to \ker\phi \to V \xrightarrow{\phi} \operatorname{im}\phi \to 0.$$

Since $\operatorname{im}\phi$ is vector space it is a free module so there exists a splitting map $h : \operatorname{im}\phi \to V$ such that $\phi \circ h = \mathrm{id}_{\operatorname{im}\phi}$ i.e. the restriction of $\phi$ to $\operatorname{im} h$ is surjective onto $\operatorname{im}\phi$. Now since there is a splitting map we get that

$$V = \ker\phi \oplus \operatorname{im} h$$

so $\ker\phi$ and $\operatorname{im} h$ are disjoint and thus $\phi \restriction \operatorname{im} h$ is in fact an isomorphism so must be non-singular. Of course $\phi \restriction \ker\phi$ is the zero map so nilpotent in a trivial sense.

**Problem:** The subspaces above might not be invariant.

**Proof**

Consider the chains

$$\ker \phi \subseteq \ker \phi^2 \subseteq \cdots$$
$$\operatorname{im} \phi \supseteq \operatorname{im} \phi^2 \supseteq \cdots$$

since the space is finite dimensional these chains must converge. Pick $k$ large enough so that both chains converge. That $\operatorname{im} \phi^k = \operatorname{im} \phi^{k+1}$ and $\ker \phi^k = \ker \phi^{k+1}$. We claim that $\operatorname{im} \phi^k \cap \ker \phi^k = \{e\}$. Indeed suppose $v \in \operatorname{im} \phi^k \cap \ker \phi^k$. Then $v = \phi^k(u)$ and since $\phi^k(v) = 0$ we see that

$$\phi^k(v) = \phi^{2k}(u) = 0$$

thus $u \in \ker \phi^{2k} = \ker \phi^k$ so that $\phi^k(u) = v = 0$. Then we have a direct sum

$$\operatorname{im} \phi^k \oplus \ker \phi^k \subseteq V$$

thus by rank nullity

$$\dim\left(\operatorname{im} \phi^k \oplus \ker \phi^k\right) = \dim \operatorname{im} \phi^k + \dim \ker \phi^k = \dim V$$

so the direct sum is the entire space as desired.

Now since the chains have converged both of these are invariant subspaces for $\phi$. That is

$$\phi\left(\operatorname{im} \phi^k\right) = \operatorname{im} \phi^{k+1} = \operatorname{im} \phi^k$$

and if $v \in \ker \phi^k$ then $\phi(v) \in \ker \phi^k$ since $\phi^k(\phi(v)) = \phi\left(\phi^k(v)\right) = \phi(0) = 0$.

Then $\phi \restriction \operatorname{im} \phi^k$ is an isomorphism since it is surjective and disjoint from the kernel of $\phi^k$ which contains the kernel of $\phi$.

$\phi \restriction \ker \phi^k$ is nilpotent since $\phi^k$ is zero on the subspace.

**Remark.** For the argument about chains note that the containments are proper iff the dimension actually drops since if the dimension doesn't drop then the subspace contains a basis for the original space so they must be equal. Alternatively there is a more general theory of Artinian modules and vector spaces are that. That $\phi$ is an isomorphism can also be justified from the fact that any surjective operator on a finite dimensional space must be injective. This is because rank nullity yields the dimension of the kernel must be 0.

□

## Problem 5

(10 points) Let $A$ be an $n \times n$ matrix.

**(a)** Suppose that $v$ is a column vector such that the set $\{v, Av, \ldots, A^{n-1}v\}$ is linearly independent. Show that any matrix $B$ that commutes with $A$ is a polynomial in $A$.

**(b)** Show that there exists a column vector $v$ such that the set $\{v, Av, \ldots, A^{n-1}v\}$ is linearly independent if and only if the characteristic polynomial of $A$ equals the minimal polynomial of $A$.

*Proof.* **(a)** Since $\{v, Av, \ldots, A^{n-1}v\}$ is linearly independent and has $n$ elements, it's a basis for $F^n$, i.e.

$$F^n = \text{span}\{v, Av, \ldots, A^{n-1}v\}.$$

Suppose $AB = BA$. Let $u \in F^n$ be arbitrary and write $u = \sum_{i=0}^{n-1} a_i A^i v$. Then

$$B(u) = B\left(\sum_{i=0}^{n-1} a_i A^i v\right) = \sum_{i=0}^{n-1} a_i A^i (Bv).$$

Say $Bv = \sum_{j=0}^{n-1} b_j A^j v$. Then

$$\begin{aligned} B(u) &= \sum_{i=0}^{n-1} a_i A^i \left(\sum_{j=0}^{n-1} b_j A^j v\right) = \sum_{i=0}^{n-1}\sum_{j=0}^{n-1} a_i b_j A^i A^j v \\ &= \sum_{j=0}^{n-1}\sum_{i=0}^{n-1} b_j A^j (a_i A^i v) = \sum_{j=0}^{n-1} b_j A^j u. \end{aligned}$$

So if $f(x) = \sum_{j=0}^{n-1} b_j x^j$, then $B = f(A)$.

**(b)** For the converse ($\implies$ direction) also consider the contrapositive. Suppose $m_A \neq \chi_A$. Then since $m_A \mid \chi_A$ and they are both monic this can only happen if $\deg m_A < \deg \chi_A$. Then $\deg m_A \leq n-1$ so we may write $m_A(x) = \sum_{i=0}^{n-1} m_i x^i$. Now $m_A(A) = 0$ so for any $v \in F^n$ we have

$$m_A(A)\, v = m_0 v + m_1 Av + \cdots + m_{n-1} A^{n-1} v = 0$$

so $\mathcal{B}_v$ is not independent for any $v$.

For the $\impliedby$ direction view $F^n$ as an $F[x]$-module via $x \cdot w := Aw$; call this module $(F^n, A)$. By the structure theorem for finitely generated torsion modules over the PID $F[x]$, there is an invariant factor decomposition

$$(F^n, A) \cong \bigoplus_{i=1}^{m} \frac{F[x]}{(d_i(x))}$$

with $d_1(x) \mid d_2(x) \mid \cdots \mid d_m(x)$, where the largest invariant factor $d_m(x)$ is the minimal polynomial $m_A(x)$, and the product $d_1(x) \cdots d_m(x)$ is the characteristic polynomial $\chi_A(x)$.

Suppose $m_A = \chi_A$. Comparing degrees in the product $d_1 \cdots d_m = \chi_A = m_A = d_m$ forces $m = 1$: there is only one invariant factor. So

$$(F^n, A) \cong \frac{F[x]}{(d(x))}, \quad d(x) = m_A(x) = \chi_A(x), \quad \deg d = n.$$

Let $\phi : F[x]/(d(x)) \to F^n$ be this $F[x]$-module isomorphism. As an $F$-vector space, $F[x]/(d(x))$ has basis $\{1, x, \ldots, x^{n-1}\}$ since $\deg d = n$.

Let $v = \phi(1)$. Since $\phi$ is an $F[x]$-module map and the module structure on $F^n$ is given by $A$, we have $\phi(x^i) = x^i \cdot \phi(1) = A^i v$. So

$$\phi(1), \phi(x), \ldots, \phi\left(x^{n-1}\right) = v, Av, \ldots, A^{n-1}v,$$

and since $\phi$ is an isomorphism it carries the basis $\{1, x, \ldots, x^{n-1}\}$ to a basis $\{v, Av, \ldots, A^{n-1}v\}$ of $F^n$. In particular this set is independent, as desired.

**Alternate proof of $\implies$ direction:** Consider again $F^n$ as a $F[x]$-module

$$(F^n, A) \cong \bigoplus_{i=1}^{m} \frac{F[x]}{(d_i(x))}$$

with isomorphism $\phi : (F^n, A) \to F^n$. Then suppose

$$\left\{v, Av, \ldots, A^{n-1}v\right\}$$

is a basis for $F^n$. Then $\phi^{-1}(v)$ generates $F^n$ as a $F[x]$-module i.e. it is cyclic isomorphic to $F[x]/\langle d(x)\rangle$ for some $d$. Then we can't have more then one invariant factor $m = 1$ and $\chi_A = M_A$. □

## Problem 6

(10 points) Let $R$ be a commutative ring, and let $M$ be an $R$-module. An $R$-submodule $N$ of $M$ is maximal if there is no $R$-module $P$ with $N \subsetneq P \subsetneq M$.

**(a)** Show that an $R$-submodule $N$ of $M$ is maximal iff $M/N$ is a simple $R$-module: i.e., $M/N$ is nonzero and has no proper, nonzero $R$-submodules.

**(b)** Let $M$ be a $\mathbb{Z}$-module. Show that a $\mathbb{Z}$-submodule $N$ of $M$ is maximal iff $\#M/N$ is a prime number.

**(c)** Let $M$ be the $\mathbb{Z}$-module of all roots of unity in $\mathbb{C}$ under multiplication. Show that there is no maximal $\mathbb{Z}$-submodule of $M$.

*Proof.* **(a)** Recall the lattice theorem for modules. That is submodules of $M/N$ correspond bijectively to submodules of $M$ containing $N$. Then if $N$ is maximal there are no proper submodules

of $M$ containing it so there are no proper submodules of $M/N$. Similarly if $M/N$ is simple then there must not be any proper submodules of $M$ containing $N$ so $N$ must be maximal.

**(b)** By part (a) $N$ is maximal iff $M/N$ is simple. Then it suffices to show that the non-zero simple $\mathbb{Z}$-modules are precisely $C_p$ for some prime $p$. Then suppose $M$ is a non-zero simple $\mathbb{Z}$-module. Then it has a nonzero element $v \in M$ and $\langle v \rangle \subseteq M$ is a submodule. Since $M$ is simple this submodule is not-proper and $\langle v \rangle = M$. Now suppose $M$ is infinite. Then $v$ has infinite order so $\langle 2v \rangle$ is a proper submodule. Then $M$ must be finite, and thus is a finite cyclic group of order say $m = |M|$. Then for any $d \mid m$ we know $\left\langle \frac{m}{d} v \right\rangle$ is a proper submodule. Then $M$ is simple iff it has no proper non-trivial divisors i.e. if $m$ is prime.

**(c)** More explicitly

$$M = \{z \in \mathbb{C} : z^n = 1 \text{ for some } n \in \mathbb{N}\}.$$

Suppose $N$ is maximal submodule so that $M/N$ is simple. Then this has prime order $p$. Then every coset satisfies

$$(zN)^p = z^p N = N$$

i.e. every $z \in M$ satisfies $z^p \in N$. We will show towards contradiction that $N = M$. Indeed suppose $z = e^{2\pi ik/n}$ is an $n$-th root of unity. Then $w = e^{2\pi ik/pn}$ is a $pn$-th root of unity hence $w \in M$. Then

$$z = w^p \in N$$

as desired.

**Remark.** Apparently there is some theory to this question. Something about this notion of divisible modules.

Divisible group – Wikipedia

□

## Problem 7

(10 points) Let $R$ be a commutative ring.

**(a)** Let $r \in R$. Show that the map $r\bullet : R \to R$ by $x \mapsto rx$ is an $R$-module endomorphism of $R$.

**(b)** We say that $r$ is a zero-divisor if $r\bullet$ is not injective. Show that if $r$ is a zero-divisor and $r \neq 0$, then the kernel and image of $R$ each consist of zero-divisors.

**(c)** Let $n \geq 2$ be an integer. Show: if $R$ has exactly $n$ zero-divisors, then $\#R \leq n^2$.

**(d)** Show that up to isomorphism there are exactly two commutative rings $R$ with precisely

2 zero-divisors. You may use without proof the following fact: every ring of order 4 is isomorphic to exactly one of the following: $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/2\mathbb{Z}[t]/(t^2+t+1)$, $\mathbb{Z}/2\mathbb{Z}[t]/(t^2-t)$, $\mathbb{Z}/2\mathbb{Z}[t]/(t^2)$.

*Proof.* **(a)** We check it preserves addition and scaling. Denote the map instead by $m_r(x)$. Then

$$m_r(a+b) = r(a+b) = ra + rb = m_r(a) + m_r(b)$$

and if $r' \in R$ then

$$m_r(r'a) = rr'a = r'(ra) = r'm_r(a)$$

so indeed multiplication by $r$ is an $R$-module endomorphism.

**(b)** Note that

$$\ker m_r = \{a \in R : ra = 0\}$$
$$\operatorname{im} m_r = \{ra : a \in R\}$$

Now suppose $r \neq 0$. Then if $a \in \ker m_r$ we have that

$$0 = ra = ar = m_a(r)$$

so $r \in \ker m_a$ and $m_a$ is not injective hence $a$ is a zero divisor. Now suppose $r$ is not injective so there is $x \in \ker m_r$ i.e. $rx = 0$. Then if $ra \in \operatorname{im} m_r$ we see that

$$m_{ra}(x) = rax = a(rx) = a0 = 0$$

so $ra$ is a zero divisor as well.

**(c)** Let $r \in R$ be a zero divisor. Then the 1st isomorphism theorem gives an isomorphism

$$R/\ker m_r \cong \operatorname{im} m_r.$$

Then since the images consists of zero divisors in $R$ and there are at most $n$ the image has size at most $n$. Then from our isomorphism this is precisely the number of distinct cosets of $\ker m_r$. Then also not that since $\ker m_r$ consists of zero divisors it also has size at most $n$ so all the cosets are finite with size at most $n$. Then we may write $r$ as a union of its distinct cosets

$$|R| = \sum_{i=1}^{n} |k_i + \ker m_r| = \sum_{i=1}^{n} |\ker m_r| \leq \sum_{i=1}^{n} n = n^2.$$

**(d)** Form part (c) $|R| \leq 4$ so we only need to check rings of size at most 4 and at least 2.

Every ring has 1 zero divisor namely 0 so we just need to consider when the ring has precisely 1 more.

**Size 2:** The ring $\{0, a\}$ where $a^2 = 2a = 0$ is a ring with precisely two zero divisors.

**Size 3:** Suppose $\{0, a, b\}$ is a ring. Since $a, b \neq 0$ we must have $a = -b$ so one is a zero divisor iff the other is and there are no rings of size 3 with two zero divisors.

**Size 4:** We consider the hint. Note that since $\mathbb{Z}/2\mathbb{Z} = \mathbb{F}_2$ is a field $\mathbb{F}_2[x]$ is a PID hence has no zero divisors. Then $\mathbb{F}_2[x]/\langle p(x)\rangle$ has zero divisors iff $p$ factors. Then we consider each polynomial.

- $t^2 + t + 1$. This is irred in $\mathbb{F}_2$ by checking for roots thus $\frac{\mathbb{F}_2[t]}{\langle t^2+t+1\rangle}$ is a field and has no zero divisors.
- $t^2 - t = t(t-1)$ factors into distinct factors so the quotient two zero divisors which is one to many.
- $t^2 = t \cdot t$ makes $t$ nilpotent hence adds only one zero divisor so this is the correct ring.

□